

affirmoTM tutorial 6: walkthrough

One run followed in detail: patients, costs, QALYs and how the arms behave over time

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1 About this tutorial

This tutorial follows the example inputs through the model: what goes in, what each function produces, and how the two arms behave over time.

- **Inputs:** each one is read with the package's own reader and explored with short `data.table` commands.
- **One run, followed step by step:** a few patients are followed year by year, with their costs and QALYs worked out by hand.
- **Over time:** the cohort is followed across the horizon in both arms.
- **The full model:** `tm_run()` at the end.

It starts from a fresh analysis folder with the example input files (see tutorial 1):

```
library(affirmoTM)
```

```
tm_example_inputs()
```

2 The inputs

2.1 Settings

```
settings <- tm_read_settings()
str(settings)
```

```
List of 9
 $ strata          : chr [1:3] "agegrp" "sex" "mm_cluster"
 $ seed            : int 2026
 $ horizon         : int 10
 $ age_limit_death_state: chr "CurrentDeath"
 $ discount_rate   : num 0.03
 $ n_runs          : int 100
 $ events          :List of 3
  ..$ ischaemic_stroke : chr [1:5] "AIS" "AIS_AB" "PB_AIS" "AIS_Death" ...
  ..$ haemorrhagic_stroke: chr [1:5] "AHS" "AHS_AB" "PB_AHS" "AHS_Death" ...
  ..$ major_bleed      : chr [1:7] "AB" "AIS_AB" "AHS_AB" "PS_AB" ...
 $ currency        : chr "€"
 $ wtp             : int [1:3] 20000 30000 50000
```

2.2 Transition matrices

`tm_read_matrices()` reads both matrices and checks them. Each one is a `tm_matrix`; printing it gives an overview:

```
matrices <- tm_read_matrices(settings)
matrices$usual_care
```

```
<tm_matrix> transition matrix (source: user)
 Strata:      agegrp (9), sex (2), mm_cluster (7) -> 126 strata
 Living states: 13 - NoEvent, AB, AHS, AIS, PB, PHS, PIS, AHS_AB, AIS_AB, PB_AHS, PB_AIS, PS_AB, PS_PB
 Death states:  6 - CurrentDeath, AB_Death, AHS_Death, AIS_Death, AHS_AB_Death, AIS_AB_Death
 Years:       the same rows every year
 Draws:      none (point values)
```

The probabilities are in `probs`, one row per stratum and current state (`from`), and one column per next state:

```
usual_care <- matrices$usual_care$probs
iabc <- matrices$iabc$probs
states <- matrices$usual_care$states
death <- matrices$usual_care$death
strata <- settings$strata
age_order <- unique(usual_care$agegrp)[order(as.integer(sub("-.*", "",
  ↪ unique(usual_care$agegrp)))] # for plots
arm_colours <- c(usual_care = "grey35", iabc = "#2a78d6")
dim(usual_care)
```

```
[1] 2394  23
```

```
usual_care[1:3, 1:10]
```

	agegrp	sex	mm_cluster	from	NoEvent	AB	AHS	AIS
	<char>	<char>	<char>	<char>	<num>	<num>	<num>	<num>
1:	70-74	men	Unspecified	NoEvent	0.9693759	0.004641617	0.002450905	0.006487908
2:	75-79	men	Unspecified	NoEvent	0.9571437	0.005880312	0.002995982	0.008038839
3:	65-69	men	Unspecified	NoEvent	0.9753106	0.003494647	0.001866399	0.005250628
	PB	PHS						
	<num>	<num>						
1:	0	0						
2:	0	0						
3:	0	0						

Every row sums to 1, in both matrices:

```
usual_care[, .(min = min(rowSums(.SD)), max = max(rowSums(.SD))), .SDcols = states]
```

	min	max
	<num>	<num>
1:	1	1

```
iabc[, .(min = min(rowSums(.SD)), max = max(rowSums(.SD))), .SDcols = states]
```

	min	max
	<num>	<num>
1:	1	1

The matrices are easier to explore in long form, one row per move, with both arms side by side:

```
long_usual_care <- melt(usual_care, id.vars = c(strata, "from"), variable.name = "to",
  ↪ value.name = "p",
  variable.factor = FALSE)
long_iabc <- melt(iabc, id.vars = c(strata, "from"), variable.name = "to", value.name = "p",
  ↪ variable.factor = FALSE)
moves <- long_usual_care[long_iabc, on = c(strata, "from", "to"),
  .(agegrp, sex, mm_cluster, from, to, usual_care = p, iabc = i.p)]
moves
```

	agegrp	sex	mm_cluster	from	to	usual_care	iabc
	<char>	<char>	<char>	<char>	<char>	<num>	<num>
1:	70-74	men	Unspecified	NoEvent	NoEvent	0.9693759	0.9707387
2:	75-79	men	Unspecified	NoEvent	NoEvent	0.9571437	0.9588437
3:	65-69	men	Unspecified	NoEvent	NoEvent	0.9753106	0.9763731
4:	80-84	men	Unspecified	NoEvent	NoEvent	0.9348558	0.9370792
5:	85-89	men	Unspecified	NoEvent	NoEvent	0.8995875	0.9022077

45482:	105-109	women	Metabolic	AB_Death	AIS_AB_Death	0.0000000	0.0000000
45483:	105-109	women	Metabolic	AHS_Death	AIS_AB_Death	0.0000000	0.0000000
45484:	105-109	women	Metabolic	AIS_Death	AIS_AB_Death	0.0000000	0.0000000
45485:	105-109	women	Metabolic	AHS_AB_Death	AIS_AB_Death	0.0000000	0.0000000
45486:	105-109	women	Metabolic	AIS_AB_Death	AIS_AB_Death	1.0000000	1.0000000

A death state's rows always go back to the same state. In every stratum, the only move out of a death state is to itself, with probability 1:

```
moves[from %in% death & usual_care > 0, .(rows = .N, probability = unique(usual_care)), by =
  ↪ .(from, to)]
```

	from	to	rows	probability
	<char>	<char>	<int>	<num>
1:	CurrentDeath	CurrentDeath	126	1
2:	AB_Death	AB_Death	126	1
3:	AHS_Death	AHS_Death	126	1
4:	AIS_Death	AIS_Death	126	1
5:	AHS_AB_Death	AHS_AB_Death	126	1
6:	AIS_AB_Death	AIS_AB_Death	126	1

The moves that are possible from each state, i.e. with a probability above 0 in at least one stratum:

```
knitr::kable(moves[usual_care > 0, .(possible_next_states = paste(unique(to), collapse = ",  
↵ ")), by = from])
```

from	possible_next_states
NoEvent	NoEvent, AB, AHS, AIS, AHS_AB, AIS_AB, CurrentDeath, AB_Death, AHS_Death, AIS_Death, AHS_AB_Death, AIS_AB_Death
PB	AB, PB, PB_AHS, PB_AIS, CurrentDeath, AB_Death
PHS	AHS, PHS, PS_AB, CurrentDeath
PIS	AIS, PIS, PS_AB, CurrentDeath, AHS_Death, AIS_Death
AB	PB, PB_AHS, PB_AIS, CurrentDeath, AB_Death, AHS_Death, AIS_Death, AIS_AB_Death
AHS	PHS, AHS_AB, PS_AB, CurrentDeath, AHS_Death, AIS_Death
AIS	PIS, AIS_AB, PS_AB, CurrentDeath, AB_Death, AHS_Death, AIS_Death
PS_PB	PB_AHS, PB_AIS, PS_AB, PS_PB, CurrentDeath
AHS_AB	PS_PB, CurrentDeath
AIS_AB	PS_PB, CurrentDeath, AB_Death
PB_AIS	PS_PB, CurrentDeath
PS_AB	PS_PB, CurrentDeath
PB_AHS	PS_PB, CurrentDeath
CurrentDeath	CurrentDeath
AB_Death	AB_Death
AHS_Death	AHS_Death
AIS_Death	AIS_Death
AHS_AB_Death	AHS_AB_Death
AIS_AB_Death	AIS_AB_Death

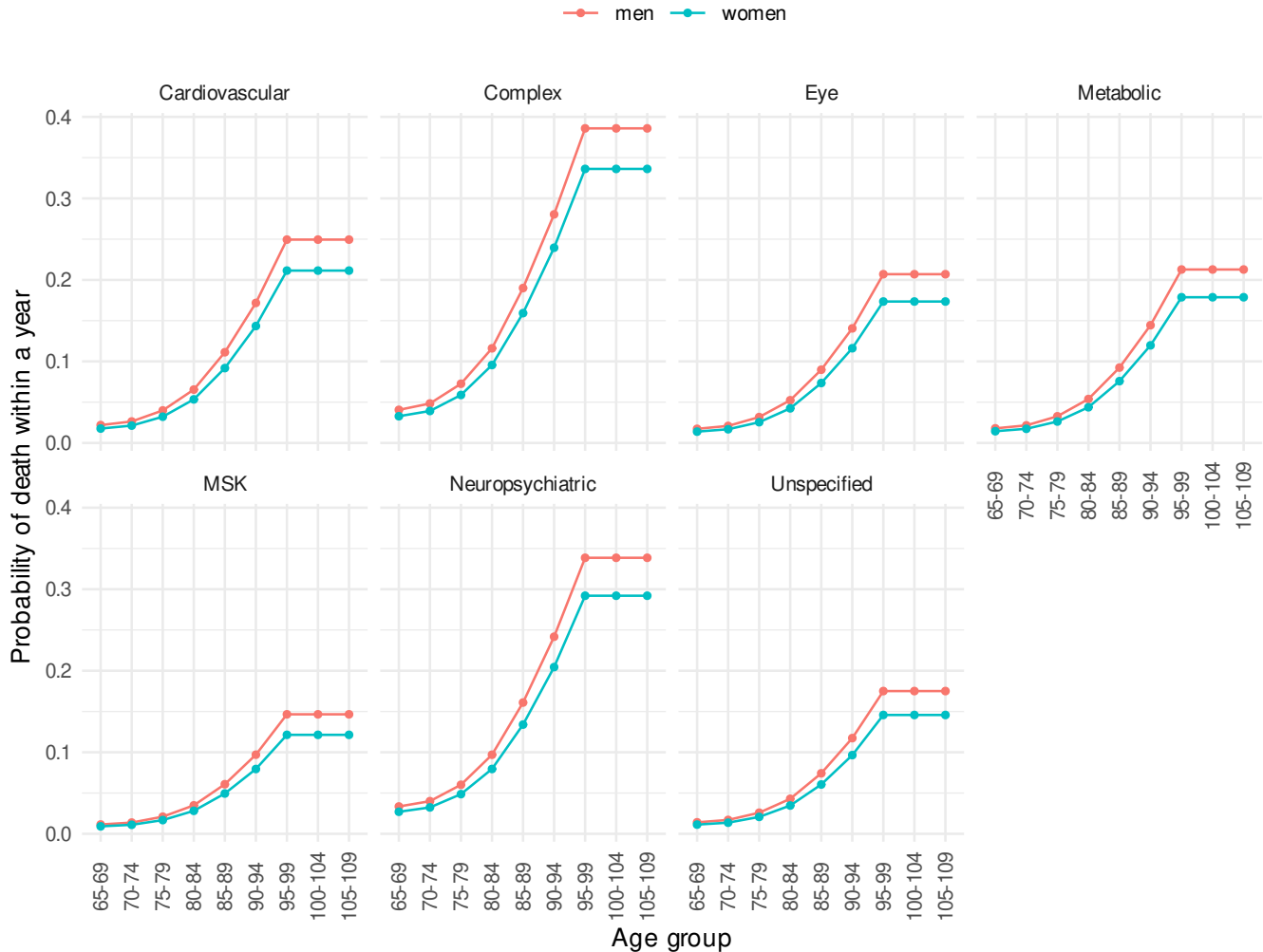
The same patient under each arm: a man aged 75–79 in the Unspecified cluster, without an event this year. Each number is the probability of being in that state next year:

```
moves[agegrp == "75-79" & sex == "men" & mm_cluster == "Unspecified" & from == "NoEvent" &  
↵ usual_care > 0,  
  .(to, usual_care, iabc, difference = iabc - usual_care)]
```

	to	usual_care	iabc	difference
	<char>	<num>	<num>	<num>
1:	NoEvent	9.571437e-01	9.588437e-01	1.699973e-03
2:	AB	5.880312e-03	5.292281e-03	-5.880312e-04
3:	AHS	2.995982e-03	2.696384e-03	-2.995982e-04
4:	AIS	8.038839e-03	7.234955e-03	-8.038839e-04
5:	AHS_AB	7.894549e-100	7.105094e-100	-7.894549e-101
6:	AIS_AB	8.459520e-05	7.613568e-05	-8.459520e-06
7:	CurrentDeath	2.364729e-02	2.364729e-02	0.000000e+00
8:	AB_Death	4.052030e-04	4.052030e-04	0.000000e+00
9:	AHS_Death	7.969846e-04	7.969846e-04	0.000000e+00
10:	AIS_Death	9.614472e-04	9.614472e-04	0.000000e+00
11:	AHS_AB_Death	1.183016e-164	1.183016e-164	0.000000e+00
12:	AIS_AB_Death	4.563169e-05	4.563169e-05	0.000000e+00

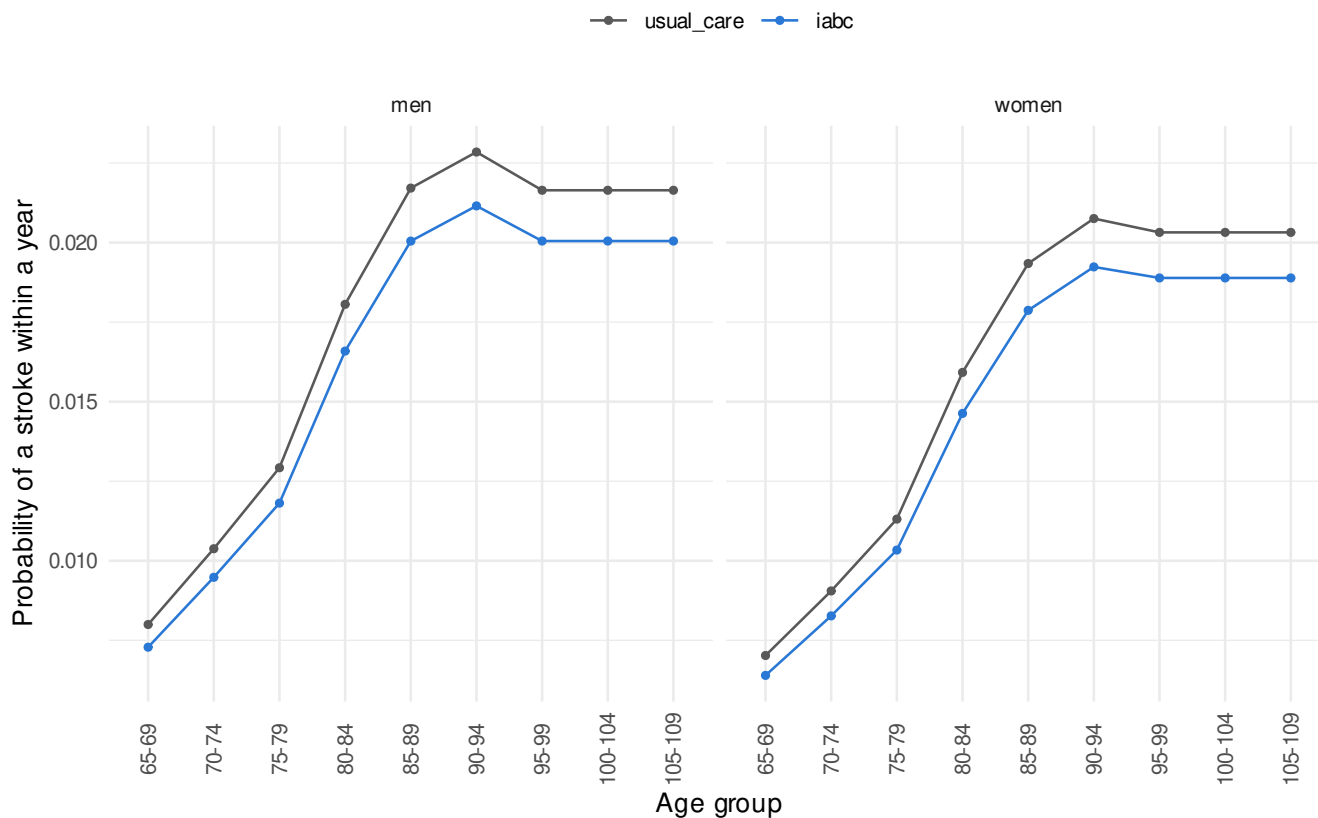
The yearly probability of death for patients without an event, by age group, sex and cluster:

```
death_by_age <- moves[from == "NoEvent" & to %in% death, .(death = sum(usual_care)), by =
  ↪ .(agegrp, sex, mm_cluster)]
ggplot(death_by_age, aes(factor(agegrp, levels = age_order), death, colour = sex, group = sex))
  ↪ +
  geom_line() + geom_point(size = 1) +
  facet_wrap(~mm_cluster, ncol = 4) +
  labs(x = "Age group", y = "Probability of death within a year", colour = NULL) +
  theme_minimal(base_size = 10) + theme(axis.text.x = element_text(angle = 90, vjust = 0.5),
  ↪ legend.position = "top")
```



The yearly probability of a stroke for patients without an event, under each arm, for the Unspecified cluster:

```
stroke_states <- unique(unlist(settings$events[c("ischaemic_stroke", "haemorrhagic_stroke")]))
stroke_by_age <- moves[from == "NoEvent" & to %in% stroke_states & mm_cluster == "Unspecified",
  .(usual_care = sum(usual_care), iabc = sum(iabc)), by = .(agegrp, sex)]
ggplot(melt(stroke_by_age, id.vars = c("agegrp", "sex"), variable.name = "arm", value.name =
  ↪ "stroke"),
  aes(factor(agegrp, levels = age_order), stroke, colour = arm, group = arm)) +
  geom_line() + geom_point(size = 1) + facet_wrap(~sex) + scale_colour_manual(values =
  ↪ arm_colours, breaks = names(arm_colours)) +
  labs(x = "Age group", y = "Probability of a stroke within a year", colour = NULL) +
  theme_minimal(base_size = 10) + theme(axis.text.x = element_text(angle = 90, vjust = 0.5),
  ↪ legend.position = "top")
```



2.3 Cohort

`tm_read_cohort()` reads the patients and works out each patient's age group from their age:

```
cohort <- tm_read_cohort(settings, matrices)
cohort
```

	id	entry_cycle	age	agegrp	sex	mm_cluster
	<int>	<int>	<int>	<char>	<char>	<char>
1:	1	0	65	65-69	men	Neuropsychiatric
2:	2	0	65	65-69	men	Neuropsychiatric
3:	3	0	65	65-69	women	Complex
4:	4	0	65	65-69	women	Eye
5:	5	0	65	65-69	women	Cardiovascular

4996:	4996	9	94	90-94	women	Unspecified
4997:	4997	9	95	95-99	women	Complex
4998:	4998	9	96	95-99	women	Eye
4999:	4999	9	97	95-99	women	Complex
5000:	5000	9	99	95-99	women	MSK

Patients enter in different cycles. Cycle 0 (2013) holds everyone already living with AF when the data start, and later cycles hold new patients:

```
cohort[, .(patients = .N, mean_age = round(mean(age), 1)), keyby = entry_cycle]
```

```
Key: <entry_cycle>
  entry_cycle patients mean_age
    <int>      <int>    <num>
1:         0         0        65
2:         0         0        65
3:         0         0        65
4:         0         0        65
5:         0         0        65
6:         0         0        65
7:         0         0        65
8:         0         0        65
9:         0         0        65
10:        9         9        94
11:        9         9        95
12:        9         9        96
13:        9         9        97
14:        9         9        99
```

1:	0	2664	78.6
2:	1	245	74.0
3:	2	272	75.1
4:	3	236	74.1
5:	4	240	74.1
6:	5	272	73.3
7:	6	249	75.3
8:	7	251	74.5
9:	8	260	74.5
10:	9	311	75.1

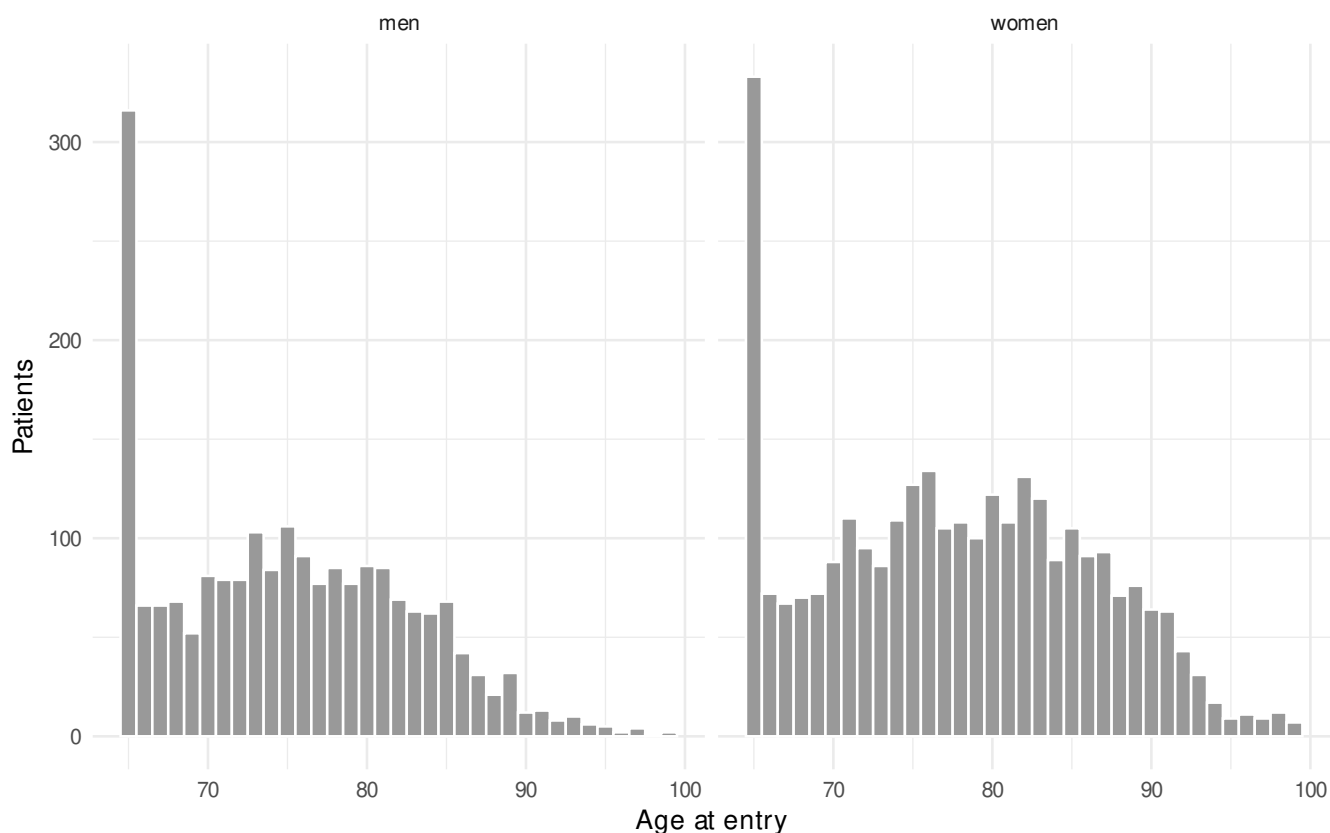
```
cohort[, .N, keyby = sex]
```

```
Key: <sex>
  sex      N
<char> <int>
1:  men 2052
2: women 2948
```

```
cohort[, .N, keyby = mm_cluster]
```

```
Key: <mm_cluster>
 mm_cluster      N
<char> <int>
1: Cardiovascular 1720
2:      Complex   362
3:      Eye       603
4:      MSK       881
5:      Metabolic  345
6: Neuropsychiatric 721
7:      Unspecified 368
```

```
ggplot(cohort, aes(age)) +
  geom_histogram(binwidth = 1, fill = "grey60", colour = "white") +
  facet_wrap(~sex) +
  labs(x = "Age at entry", y = "Patients") +
  theme_minimal(base_size = 10)
```



2.4 Entry states

`tm_read_entry_states()` reads the probability of each state in a patient's entry year, one row per stratum. Every row sums to 1:

```
entry_states <- tm_read_entry_states(settings, matrices)
entry_cols <- setdiff(names(entry_states), strata)
entry_states[, .(rows = .N, min = min(rowSums(.SD)), max = max(rowSums(.SD))), .SDcols =
  ↪ entry_cols]
```

```
  rows  min  max
<int> <num> <num>
1:    98    1    1
```

```
entry_states[agegrp == "75-79" & sex == "women" & mm_cluster == "Cardiovascular"]
```

```
  agegrp  sex  mm_cluster NoEvent  AB  AHS  AIS  PB
<char> <char>    <char>    <num> <num> <num> <num> <num>
1: 75-79 women Cardiovascular 0.9011299 0 0.004708098 0.01506591 0.005649718
    PHS      PIS AHS_AB AIS_AB PB_AHS PB_AIS PS_AB PS_PB CurrentDeath
    <num>    <num> <num> <num> <num> <num> <num> <num> <num>
1: 0.003766478 0.01506591 0 0 0 0 0 0 0.05461394
```

2.5 Costs, utilities and the intervention cost

```
costs <- tm_read_costs(settings, matrices)
utilities <- tm_read_utilities(settings, matrices)
intervention <- tm_read_intervention_cost(settings)
costs
```

	state	usual_care	iabc
	<char>	<num>	<num>
1:	NoEvent	2000	2000
2:	AB	8000	8000
3:	AHS	20000	20000
4:	AIS	15000	15000
5:	PB	3000	3000
6:	PHS	5000	5000
7:	PIS	4000	4000
8:	AHS_AB	27000	27000
9:	AIS_AB	22000	22000
10:	PB_AHS	22000	22000
11:	PB_AIS	17000	17000
12:	PS_AB	12000	12000
13:	PS_PB	6000	6000
14:	CurrentDeath	3000	3000
15:	AB_Death	7000	7000
16:	AHS_Death	15000	15000
17:	AIS_Death	12000	12000
18:	AHS_AB_Death	20000	20000
19:	AIS_AB_Death	18000	18000

```
utilities
```

	state	usual_care	iabc
	<char>	<num>	<num>
1:	NoEvent	0.75	0.75
2:	AB	0.65	0.65
3:	AHS	0.50	0.50
4:	AIS	0.55	0.55
5:	PB	0.70	0.70
6:	PHS	0.60	0.60
7:	PIS	0.65	0.65
8:	AHS_AB	0.40	0.40
9:	AIS_AB	0.45	0.45
10:	PB_AHS	0.47	0.47
11:	PB_AIS	0.52	0.52
12:	PS_AB	0.55	0.55
13:	PS_PB	0.60	0.60
14:	CurrentDeath	0.00	0.00
15:	AB_Death	0.00	0.00
16:	AHS_Death	0.00	0.00
17:	AIS_Death	0.00	0.00
18:	AHS_AB_Death	0.00	0.00
19:	AIS_AB_Death	0.00	0.00

```
intervention
```

	year	cost
	<int>	<num>
1:	1	50
2:	2	0
3:	3	0
4:	4	0
5:	5	0
6:	6	0

```
7:      7      0
8:      8      0
9:      9      0
```

- **Costs:** the cost of a year spent in each state, per arm.
- **Utilities:** the QALY weight of a year in each state, per arm. Death states count 0.
- **Intervention cost:** the cost of delivering iABC to one patient in each year after entry. It is charged to iABC patients who are alive in that year.

3 One run, step by step

`tm_run()` repeats these steps once per run. Here they are done one at a time for the first run, which uses the seed from `settings.yaml`:

```
seed <- settings$seed
seed
```

```
[1] 2026
```

3.1 Entry states

`tm_assign_entry_states()` draws each patient's state in their entry year from their row of the entry-state table. Both arms start from these states.

```
patients <- tm_assign_entry_states(cohort, entry_states, settings, seed = seed)
patients[, .N, keyby = entry_state][order(-N)]
```

	entry_state	N
	<char>	<int>
1:	NoEvent	4318
2:	CurrentDeath	420
3:	PIS	89
4:	AIS	64
5:	PB	43
6:	PHS	28
7:	AB	17
8:	AHS	16
9:	PB_AIS	2
10:	PS_AB	2
11:	AHS_AB	1

3.2 The simulation

`tm_simulate()` moves every patient through both arms one year at a time. In each later year, it draws the patient's next state from their matrix row: their arm, sex and cluster, their age group in the previous year, and their previous state. The result has one row per patient, arm and year, from the entry year up to and including the year of death:

```
sim <- tm_simulate(patients, matrices, settings, seed = seed)
setorder(sim, arm, id, cycle)
sim
```

	id	arm	cycle	age	agegrp	sex	mm_cluster	state
	<int>	<char>	<int>	<int>	<char>	<char>	<char>	<char>
1:	1	iabc	0	65	65-69	men	Neuropsychiatric	NoEvent
2:	1	iabc	1	66	65-69	men	Neuropsychiatric	NoEvent
3:	1	iabc	2	67	65-69	men	Neuropsychiatric	NoEvent
4:	1	iabc	3	68	65-69	men	Neuropsychiatric	NoEvent
5:	1	iabc	4	69	65-69	men	Neuropsychiatric	NoEvent

59551:	4996	usual_care	9	94	90-94	women	Unspecified	NoEvent
59552:	4997	usual_care	9	95	95-99	women	Complex	NoEvent
59553:	4998	usual_care	9	96	95-99	women	Eye	CurrentDeath
59554:	4999	usual_care	9	97	95-99	women	Complex	CurrentDeath
59555:	5000	usual_care	9	99	95-99	women	MSK	CurrentDeath

```
sim[, .(patients = uniqueN(id), patient_years = .N, deaths = sum(state %in% death)), by = arm]
```

	arm	patients	patient_years	deaths
	<char>	<int>	<int>	<int>
1:	iabc	5000	29782	1845
2:	usual_care	5000	29773	1846

3.3 Following five patients

Each patient's path, written as one line per arm, makes it easy to pick patients with different stories:

```
paths <- sim[, .(path = paste(state, collapse = " > "), entered = min(cycle), last_state =
  ↪ last(state)), by = .(arm, id)]
paths <- dcast(paths, id + entered ~ arm, value.var = c("path", "last_state"))
head(paths[, .(id, entered, path_usual_care)])
```

Key: <id, entered>

	id	entered
	<int>	<int>
1:	1	0
2:	2	0
3:	3	0
4:	4	0
5:	5	0
6:	6	0

	path_usual_care
	<char>
1:	NoEvent > NoEvent > NoEvent > NoEvent > NoEvent > NoEvent > NoEvent > NoEvent > NoEvent > NoEvent
2:	NoEvent > NoEvent > NoEvent > NoEvent > NoEvent > AIS > PIS > PIS > PIS > PIS
3:	NoEvent > NoEvent > NoEvent > NoEvent > NoEvent > NoEvent > NoEvent > NoEvent > NoEvent > NoEvent
4:	NoEvent > NoEvent > NoEvent > NoEvent > NoEvent > NoEvent > NoEvent > CurrentDeath
5:	NoEvent > NoEvent > NoEvent > NoEvent > NoEvent > NoEvent > NoEvent > NoEvent > NoEvent > NoEvent
6:	NoEvent > NoEvent > NoEvent > NoEvent > NoEvent > NoEvent > NoEvent > NoEvent > NoEvent > NoEvent

The five patients, chosen from these paths:

```
event_free <- paths[entered == 0 & path_usual_care == paste(rep("NoEvent", 10), collapse = " >
  ↪ "), id][1]
stroke <- paths[entered == 0 & grepl("NoEvent > AIS > PIS", path_usual_care), id][1]
died <- paths[entered == 0 & last_state_usual_care == "CurrentDeath" & grepl("^NoEvent >
  ↪ NoEvent > NoEvent", path_usual_care), id][1]
late <- paths[entered == 7 & path_usual_care == path_iabc, id][1]
differs <- paths[grepl("AIS", path_usual_care) & !grepl("AIS", path_iabc), id][1]
c(event_free = event_free, stroke = stroke, died = died, late = late, differs = differs)
```

event_free	stroke	died	late	differs
1	2	4	4179	476

Each patient's years in both arms, side by side:

A patient who stays event-free for the whole horizon:

```
dcast(sim[id == event_free], cycle + age + agegrp ~ arm, value.var = "state")
```

```
Key: <cycle, age, agegrp>
  cycle  age agegrp  iabc usual_care
  <int> <int> <char> <char>    <char>
1:     0   65 65-69 NoEvent  NoEvent
2:     1   66 65-69 NoEvent  NoEvent
3:     2   67 65-69 NoEvent  NoEvent
4:     3   68 65-69 NoEvent  NoEvent
5:     4   69 65-69 NoEvent  NoEvent
6:     5   70 70-74 NoEvent  NoEvent
7:     6   71 70-74 NoEvent  NoEvent
8:     7   72 70-74 NoEvent  NoEvent
9:     8   73 70-74 NoEvent  NoEvent
10:    9   74 70-74 NoEvent  NoEvent
```

A patient with an ischaemic stroke: acute (AIS) in the year it happens, then past (PIS) in the years after:

```
dcast(sim[id == stroke], cycle + age + agegrp ~ arm, value.var = "state")
```

```
Key: <cycle, age, agegrp>
  cycle  age agegrp  iabc usual_care
  <int> <int> <char> <char>    <char>
1:     0   65 65-69 NoEvent  NoEvent
2:     1   66 65-69 NoEvent  NoEvent
3:     2   67 65-69 NoEvent  NoEvent
4:     3   68 65-69 NoEvent  NoEvent
5:     4   69 65-69 NoEvent  NoEvent
6:     5   70 70-74    AIS    AIS
7:     6   71 70-74    PIS    PIS
8:     7   72 70-74    PIS    PIS
9:     8   73 70-74    PIS    PIS
10:    9   74 70-74    PIS    PIS
```

A patient who dies. The year of death is the last year recorded:

```
dcast(sim[id == died], cycle + age + agegrp ~ arm, value.var = "state")
```

```
Key: <cycle, age, agegrp>
  cycle  age agegrp  iabc  usual_care
  <int> <int> <char>    <char>    <char>
1:     0   65 65-69    NoEvent    NoEvent
2:     1   66 65-69    NoEvent    NoEvent
3:     2   67 65-69    NoEvent    NoEvent
4:     3   68 65-69    NoEvent    NoEvent
5:     4   69 65-69    NoEvent    NoEvent
6:     5   70 70-74    NoEvent    NoEvent
7:     6   71 70-74 CurrentDeath CurrentDeath
```

A patient who enters late, in cycle 7 (2020), and is followed until the end of the horizon:

```
dcast(sim[id == late], cycle + age + agegrp ~ arm, value.var = "state")
```

```
Key: <cycle, age, agegrp>
  cycle age agegrp iabc usual_care
  <int> <int> <char> <char> <char>
1:    7   65 65-69 NoEvent NoEvent
2:    8   66 65-69 NoEvent NoEvent
3:    9   67 65-69 NoEvent NoEvent
```

A patient whose path differs between the arms. Both arms use the same random number for the same patient and year, so the paths stay together until, in one year, the same number lands on a different state in each arm's row:

```
dcast(sim[id == differs], cycle + age + agegrp ~ arm, value.var = "state")
```

```
Key: <cycle, age, agegrp>
  cycle age agegrp iabc usual_care
  <int> <int> <char> <char> <char>
1:    0   71 70-74 NoEvent NoEvent
2:    1   72 70-74 NoEvent NoEvent
3:    2   73 70-74 NoEvent NoEvent
4:    3   74 70-74 NoEvent NoEvent
5:    4   75 75-79 NoEvent NoEvent
6:    5   76 75-79 NoEvent NoEvent
7:    6   77 75-79 AHS AIS
8:    7   78 75-79 PHS PIS
9:    8   79 75-79 PHS PIS
10:   9   80 80-84 PHS PIS
```

3.4 Their costs and QALYs, by hand

`tm_add_outcomes()` gives each patient-year its cost, intervention cost, QALY and discount factor:

```
out <- tm_add_outcomes(sim, costs, utilities, intervention, death, settings)
out[id == differs, .(arm, cycle, state, cost, intervention_cost, qaly, discount)]
```

	arm	cycle	state	cost	intervention_cost	qaly	discount
	<char>	<int>	<char>	<num>	<num>	<num>	<num>
1:	iabc	0	NoEvent	2000	0	0.75	1.0000000
2:	iabc	1	NoEvent	2050	50	0.75	0.9708738
3:	iabc	2	NoEvent	2000	0	0.75	0.9425959
4:	iabc	3	NoEvent	2000	0	0.75	0.9151417
5:	iabc	4	NoEvent	2000	0	0.75	0.8884870
6:	iabc	5	NoEvent	2000	0	0.75	0.8626088
7:	iabc	6	AHS	20000	0	0.50	0.8374843
8:	iabc	7	PHS	5000	0	0.60	0.8130915
9:	iabc	8	PHS	5000	0	0.60	0.7894092
10:	iabc	9	PHS	5000	0	0.60	0.7664167
11:	usual_care	0	NoEvent	2000	0	0.75	1.0000000
12:	usual_care	1	NoEvent	2000	0	0.75	0.9708738
13:	usual_care	2	NoEvent	2000	0	0.75	0.9425959
14:	usual_care	3	NoEvent	2000	0	0.75	0.9151417
15:	usual_care	4	NoEvent	2000	0	0.75	0.8884870
16:	usual_care	5	NoEvent	2000	0	0.75	0.8626088
17:	usual_care	6	AIS	15000	0	0.55	0.8374843
18:	usual_care	7	PIS	4000	0	0.65	0.8130915
19:	usual_care	8	PIS	4000	0	0.65	0.7894092
20:	usual_care	9	PIS	4000	0	0.65	0.7664167

The same numbers worked out by hand for the five patients, from the input tables:

- **Values:** in a patient's entry year both arms take the usual-care cost and utility; from the next year, each arm's own.
- **Intervention cost:** iABC patients pay the intervention cost of that year since entry, for each year after entry that they are alive.
- **Discount factor:** a year in cycle t is discounted by $1 / (1 + \text{rate})^t$.

```
costs_long <- melt(costs, id.vars = "state", variable.name = "arm", value.name = "state_cost",
  ↪ variable.factor = FALSE)
utilities_long <- melt(utilities, id.vars = "state", variable.name = "arm", value.name =
  ↪ "utility",
                        variable.factor = FALSE)
followed <- c(event_free, stroke, died, late, differs)
by_hand <- out[id %in% followed, .(id, arm, cycle, state, cost, intervention_cost, qaly,
  ↪ discount)]
by_hand[, years_after_entry := cycle - min(cycle), by = id]
by_hand[, value_arm := fifelse(years_after_entry == 0, "usual_care", arm)]
by_hand[costs_long, on = .(state, value_arm = arm), state_cost := i.state_cost]
by_hand[utilities_long, on = .(state, value_arm = arm), utility := i.utility]
by_hand[, programme := fifelse(arm == "iabc" & years_after_entry > 0 & !state %in% death,
  ↪ intervention$cost[match(years_after_entry, intervention$year)],
  ↪ 0)]
by_hand[, `:=`(cost_by_hand = state_cost + programme, discount_by_hand = 1 / (1 +
  ↪ settings$discount_rate)^cycle)]
by_hand[id == differs, .(arm, cycle, state, state_cost, programme, cost_by_hand, cost)]
```

	arm	cycle	state	state_cost	programme	cost_by_hand	cost
	<char>	<int>	<char>	<num>	<num>	<num>	<num>
1:	iabc	0	NoEvent	2000	0	2000	2000
2:	iabc	1	NoEvent	2000	50	2050	2050
3:	iabc	2	NoEvent	2000	0	2000	2000
4:	iabc	3	NoEvent	2000	0	2000	2000
5:	iabc	4	NoEvent	2000	0	2000	2000
6:	iabc	5	NoEvent	2000	0	2000	2000
7:	iabc	6	AHS	20000	0	20000	20000
8:	iabc	7	PHS	5000	0	5000	5000
9:	iabc	8	PHS	5000	0	5000	5000
10:	iabc	9	PHS	5000	0	5000	5000
11:	usual_care	0	NoEvent	2000	0	2000	2000
12:	usual_care	1	NoEvent	2000	0	2000	2000
13:	usual_care	2	NoEvent	2000	0	2000	2000
14:	usual_care	3	NoEvent	2000	0	2000	2000
15:	usual_care	4	NoEvent	2000	0	2000	2000
16:	usual_care	5	NoEvent	2000	0	2000	2000
17:	usual_care	6	AIS	15000	0	15000	15000
18:	usual_care	7	PIS	4000	0	4000	4000
19:	usual_care	8	PIS	4000	0	4000	4000
20:	usual_care	9	PIS	4000	0	4000	4000

```
by_hand[id == differs, .(arm, cycle, state, utility, qaly, discount_by_hand =
  ↪ round(discount_by_hand, 4),
                        discount = round(discount, 4))]
```

	arm	cycle	state	utility	qaly	discount_by_hand	discount
	<char>	<int>	<char>	<num>	<num>	<num>	<num>
1:	iabc	0	NoEvent	0.75	0.75	1.0000	1.0000
2:	iabc	1	NoEvent	0.75	0.75	0.9709	0.9709
3:	iabc	2	NoEvent	0.75	0.75	0.9426	0.9426

4:	iabc	3	NoEvent	0.75	0.75	0.9151	0.9151
5:	iabc	4	NoEvent	0.75	0.75	0.8885	0.8885
6:	iabc	5	NoEvent	0.75	0.75	0.8626	0.8626
7:	iabc	6	AHS	0.50	0.50	0.8375	0.8375
8:	iabc	7	PHS	0.60	0.60	0.8131	0.8131
9:	iabc	8	PHS	0.60	0.60	0.7894	0.7894
10:	iabc	9	PHS	0.60	0.60	0.7664	0.7664
11:	usual_care	0	NoEvent	0.75	0.75	1.0000	1.0000
12:	usual_care	1	NoEvent	0.75	0.75	0.9709	0.9709
13:	usual_care	2	NoEvent	0.75	0.75	0.9426	0.9426
14:	usual_care	3	NoEvent	0.75	0.75	0.9151	0.9151
15:	usual_care	4	NoEvent	0.75	0.75	0.8885	0.8885
16:	usual_care	5	NoEvent	0.75	0.75	0.8626	0.8626
17:	usual_care	6	AIS	0.55	0.55	0.8375	0.8375
18:	usual_care	7	PIS	0.65	0.65	0.8131	0.8131
19:	usual_care	8	PIS	0.65	0.65	0.7894	0.7894
20:	usual_care	9	PIS	0.65	0.65	0.7664	0.7664

arm	cycle	state	utility	qaly	discount_by_hand	discount
<char>	<int>	<char>	<num>	<num>	<num>	<num>

Every year of the five patients, compared with the package's numbers:

```
by_hand[, .(years = .N, same_cost = all(cost == cost_by_hand), same_intervention_cost =
  ↪ all(intervention_cost == programme),
  same_qaly = all(qaly == utility), same_discount = all(abs(discount -
  ↪ discount_by_hand) < 1e-12)), by = id]
```

	id	years	same_cost	same_intervention_cost	same_qaly	same_discount
	<int>	<int>	<lgcl>	<lgcl>	<lgcl>	<lgcl>
1:	1	20	TRUE	TRUE	TRUE	TRUE
2:	2	20	TRUE	TRUE	TRUE	TRUE
3:	4	14	TRUE	TRUE	TRUE	TRUE
4:	476	20	TRUE	TRUE	TRUE	TRUE
5:	4179	6	TRUE	TRUE	TRUE	TRUE

Each patient's discounted total in each arm, which is what the arm totals add up:

```
dcast(by_hand[, .(cost = sum(cost_by_hand * discount_by_hand), qalys = sum(utility *
  ↪ discount_by_hand)), by = .(id, arm)],
  id ~ arm, value.var = c("cost", "qalys"))
```

Key: <id>

	id	cost_iabc	cost_usual_care	qalys_iabc	qalys_usual_care
	<int>	<num>	<num>	<num>	<num>
1:	1	17620.762	17572.218	6.589582	6.589582
2:	2	35247.479	35198.936	6.096420	6.096420
3:	4	13720.411	13671.867	4.184780	4.184780
4:	476	39802.231	33197.348	6.024873	6.185193
5:	4179	4777.305	4737.835	1.776688	1.776688

4 How the model behaves over time

4.1 States over time

To see the cohort move through the states, the next plots follow the 2,664 patients who enter in cycle 0, so later entrants don't blur the picture. The 19 states are grouped into five:

```

state_groups <- data.table(state = states)
state_groups[, group := fcase(state == "NoEvent", "No event",
                             state %in% death, "Dead",
                             state %in% c("AIS", "AHS", "AIS_AB", "AHS_AB", "PB_AIS",
                             ↪ "PB_AHS"), "Stroke this year",
                             state %in% c("AB", "PS_AB"), "Bleed this year",
                             default = "Past stroke or bleed")]
state_groups[, .(states = paste(state, collapse = ", ")), by = group]

```

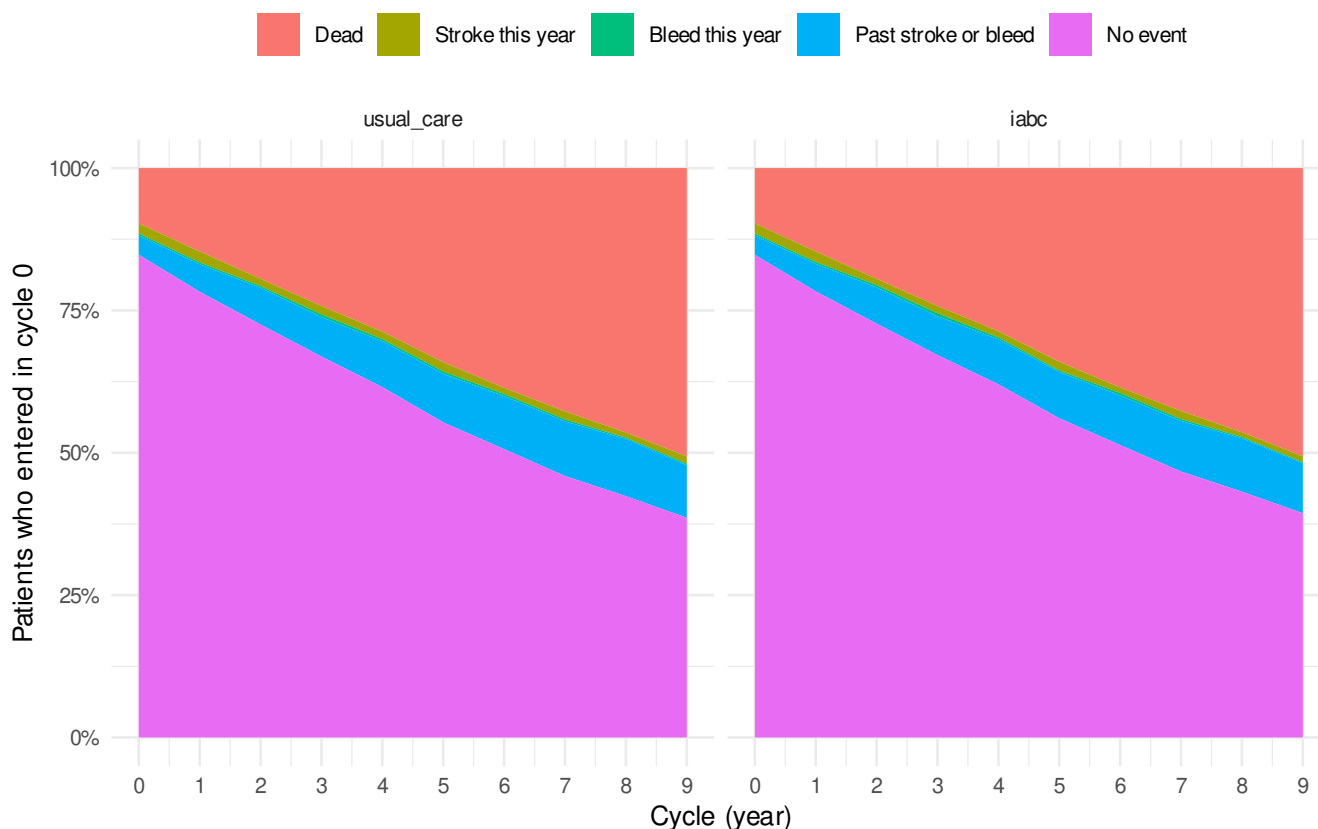
	group <char>	states <char>
1:	No event	NoEvent
2:	Bleed this year	AB, PS_AB
3:	Stroke this year	AHS, AIS, AHS_AB, AIS_AB, PB_AHS, PB_AIS
4:	Past stroke or bleed	PB, PHS, PIS, PS_PB
5:	Dead	CurrentDeath, AB_Death, AHS_Death, AIS_Death, AHS_AB_Death, AIS_AB_Death

Each year, the share of these patients in each group. A patient who has died stays counted as dead in later years. The patients already dead in cycle 0 are those whose entry state is a death state: they died in their first AF year. Both arms start from the same entry states, so they are the same in both arms:

```

first_entrants <- cohort[entry_cycle == 0, id]
yearly <- sim[id %in% first_entrants][state_groups, on = "state", nomatch = NULL][, .N, by =
↪ .(arm, cycle, group)]
# Years with nobody in a group count 0.
yearly <- yearly[CJ(arm = c("usual_care", "iabc"), cycle = 0:(settings$horizon - 1), group =
↪ unique(state_groups$group)),
on = .(arm, cycle, group)][is.na(N), N := 0L]
alive <- yearly[group != "Dead"]
dead <- yearly[group == "Dead"][order(cycle), .(cycle, N = cumsum(N), group = "Dead"), by =
↪ arm]
over_time <- rbind(alive, dead)[, share := N / length(first_entrants)]
over_time[, `:=`(group = factor(group, levels = c("Dead", "Stroke this year", "Bleed this
↪ year", "Past stroke or bleed",
                             "No event"))), arm = factor(arm, levels =
↪ c("usual_care", "iabc")))]
ggplot(over_time, aes(cycle, share, fill = group)) +
  geom_area(stat = "identity") + facet_wrap(~arm) +
  scale_x_continuous(breaks = 0:9) + scale_y_continuous(labels = scales::percent) +
  labs(x = "Cycle (year)", y = "Patients who entered in cycle 0", fill = NULL) +
  theme_minimal(base_size = 10) + theme(legend.position = "top")

```



```
dcast(over_time[cycle %in% c(0, 5, 9)], arm + group ~ paste("cycle", cycle), value.var =
  ↪ "share")
```

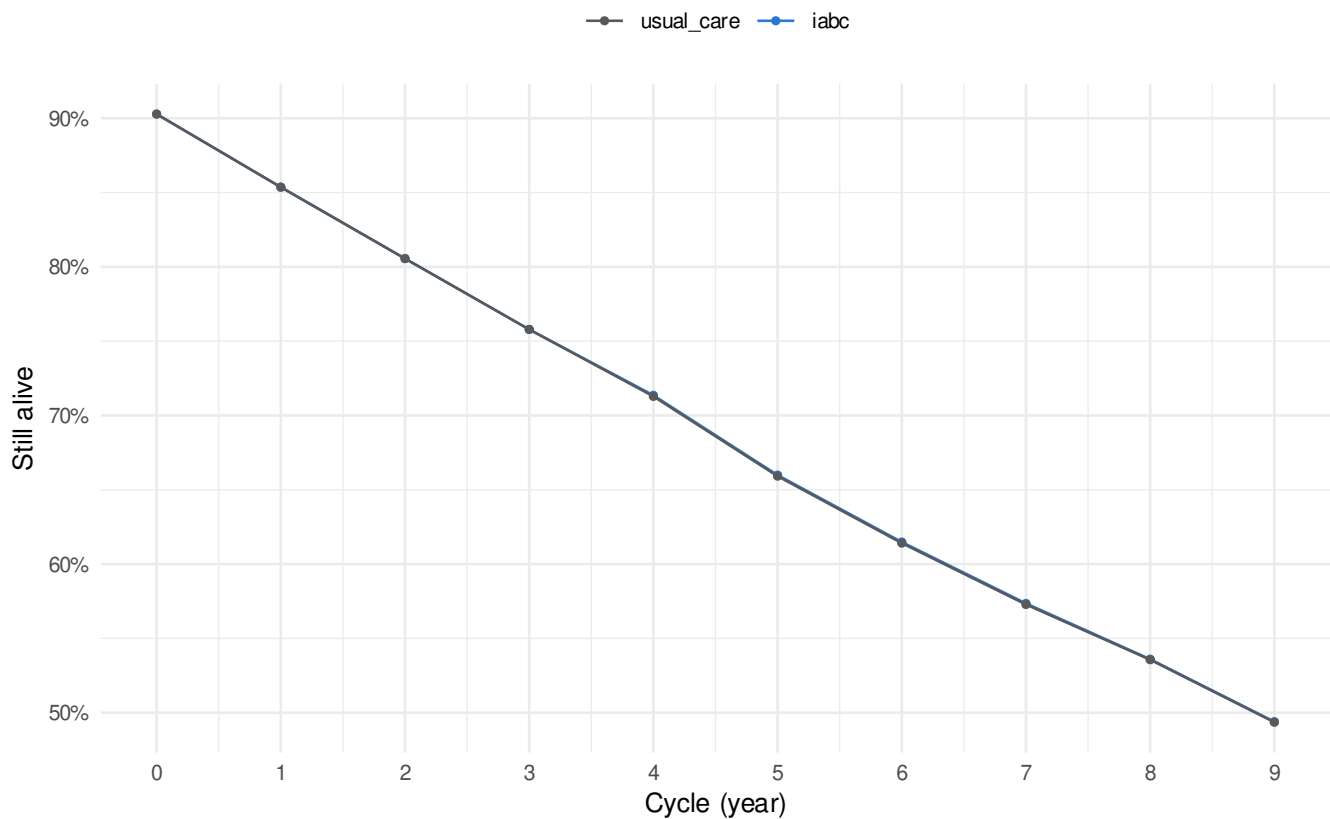
Key: <arm, group>

	arm	group	cycle 0	cycle 5	cycle 9
	<fctr>	<fctr>	<num>	<num>	<num>
1:	usual_care	Dead	0.097222222	0.340840841	0.506381381
2:	usual_care	Stroke this year	0.017642643	0.016141141	0.012762763
3:	usual_care	Bleed this year	0.003378378	0.004129129	0.002627628
4:	usual_care	Past stroke or bleed	0.034159159	0.085585586	0.092342342
5:	usual_care	No event	0.847597598	0.553303303	0.385885886
6:	iabc	Dead	0.097222222	0.340090090	0.506006006
7:	iabc	Stroke this year	0.017642643	0.015390390	0.010510511
8:	iabc	Bleed this year	0.003378378	0.002627628	0.001876877
9:	iabc	Past stroke or bleed	0.034159159	0.081456456	0.087837838
10:	iabc	No event	0.847597598	0.560435435	0.393768769

4.2 Survival

The share of these patients still alive at the end of each year, and the deaths each year in each arm. The two lines lie almost on top of each other: in this run, iABC changes the number of deaths among these 2,664 patients by only a few, as the table shows:

```
survival <- dead[, .(arm, cycle, alive = 1 - N / length(first_entrants))]
ggplot(survival, aes(cycle, alive, colour = arm)) +
  geom_line() + geom_point(size = 1) + scale_colour_manual(values = arm_colours, breaks =
    ↪ names(arm_colours)) +
  scale_x_continuous(breaks = 0:9) + scale_y_continuous(labels = scales::percent) +
  labs(x = "Cycle (year)", y = "Still alive", colour = NULL) +
  theme_minimal(base_size = 10) + theme(legend.position = "top")
```



```
deaths_by_year <- dcast(yearly[group == "Dead"], cycle ~ arm, value.var = "N")
deaths_by_year[, `:=`(difference = iabc - usual_care, cumulative_difference = cumsum(iabc -
  ↪ usual_care))]
deaths_by_year
```

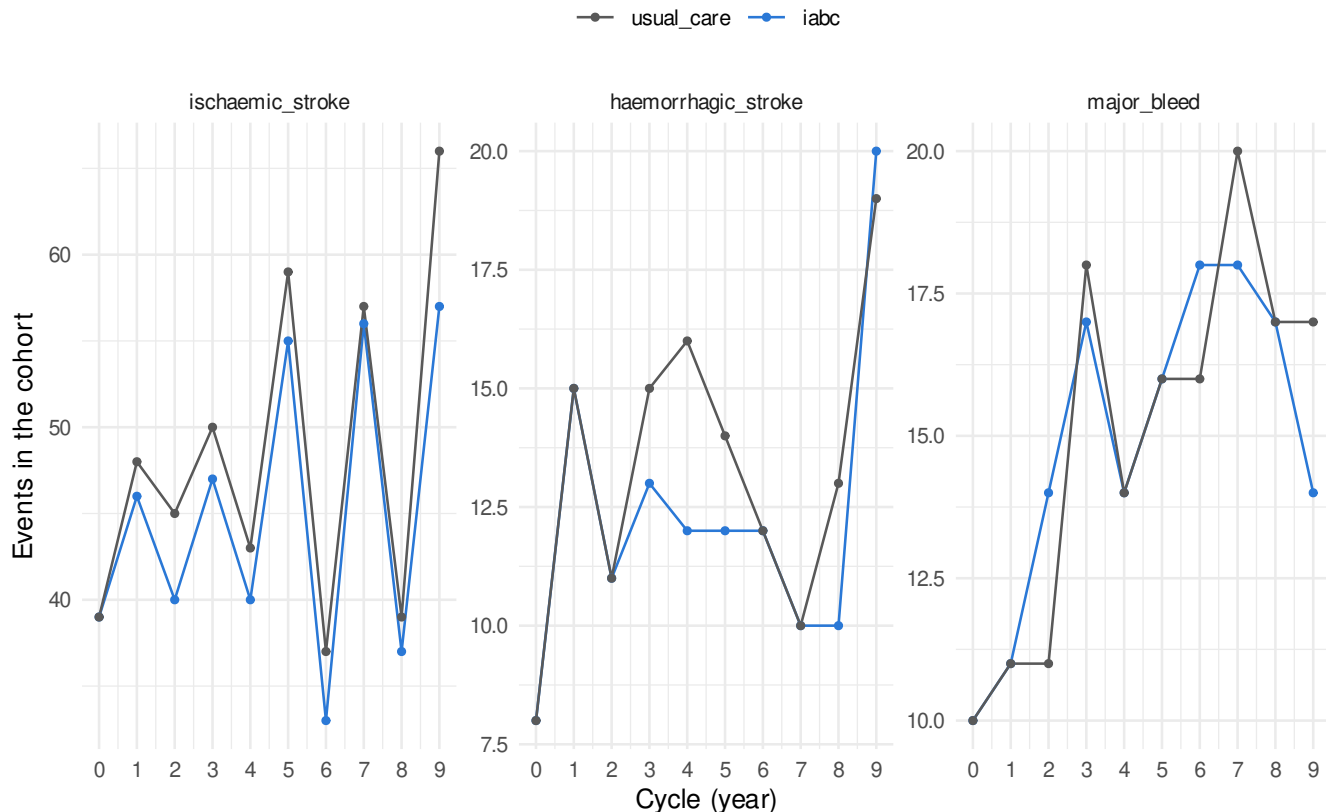
```
Key: <cycle>
  cycle  iabc usual_care difference cumulative_difference
  <int> <int>    <int>    <int>          <int>
1:     0   259      259         0                0
2:     1   131      131         0                0
3:     2   128      128         0                0
4:     3   127      127         0                0
5:     4   118      120        -2               -2
6:     5   143      143         0               -2
7:     6   120      120         0               -2
8:     7   110      110         0               -2
9:     8   100       99         1              -1
10:    9   112      112         0              -1
```

4.3 Strokes and bleeds per year

The events counted in the results (events in `settings.yaml`), each year, for the whole cohort. Each year a patient spends in one of an event's states counts as one event:

```
event_states <- setnames(as.data.table(stack(lapply(settings$events, unlist))), c("state",
  ↪ "event"))
events_by_year <- sim[event_states, on = "state", allow.cartesian = TRUE, nomatch = NULL][, .N,
  ↪ by = .(arm, cycle, event)]
ggplot(events_by_year, aes(cycle, N, colour = arm)) +
  geom_line() + geom_point(size = 1) + facet_wrap(~event, scales = "free_y") +
  ↪ scale_colour_manual(values = arm_colours, breaks = names(arm_colours)) +
```

```
scale_x_continuous(breaks = 0:9) +
labs(x = "Cycle (year)", y = "Events in the cohort", colour = NULL) +
theme_minimal(base_size = 10) + theme(legend.position = "top")
```



```
dcast(events_by_year, event ~ arm, value.var = "N", fun.aggregate = sum)[, difference := iabc
  - usual_care][,]
```

Key: <event>

	event <fctr>	iabc <int>	usual_care <int>	difference <int>
1:	ischaemic_stroke	450	483	-33
2:	haemorrhagic_stroke	123	133	-10
3:	major_bleed	149	150	-1

4.4 Costs and QALYs building up

The discounted cost and QALYs per patient, added up year by year, for the whole cohort. The cost is split into the intervention cost and the cost of care (the state costs):

```
n_patients <- uniqueN(out$id)
by_year <- out[, .(care_cost = sum((cost - intervention_cost) * discount), intervention_cost =
  sum(intervention_cost * discount),
  qalys = sum(qaly_disc)), by = .(arm, cycle)]
setorder(by_year, arm, cycle)
by_year[, `:=`(care_cost = cumsum(care_cost) / n_patients, intervention_cost =
  cumsum(intervention_cost) / n_patients,
  qalys = cumsum(qalys) / n_patients), by = arm]
difference <- dcast(by_year, cycle ~ arm, value.var = c("care_cost", "intervention_cost",
  "qalys"))
```

```

difference <- difference[, .(cycle, care_cost = care_cost_iabc - care_cost_usual_care,
                             intervention_cost = intervention_cost_iabc -
                             ↪ intervention_cost_usual_care,
                             total_cost = care_cost_iabc + intervention_cost_iabc -
                             ↪ care_cost_usual_care - intervention_cost_usual_care,
                             qalys = qalys_iabc - qalys_usual_care)]
difference

```

Key: <cycle>

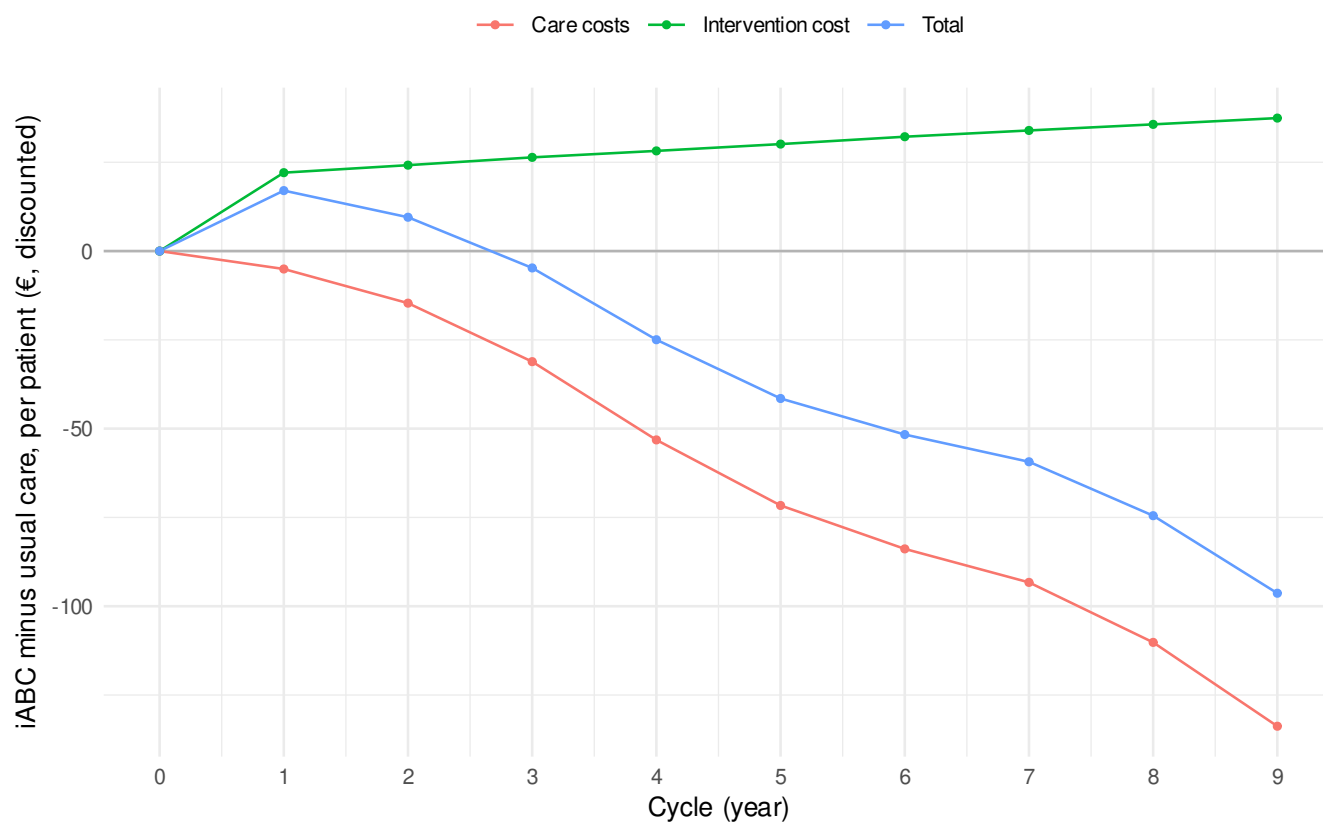
	cycle <int>	care_cost <num>	intervention_cost <num>	total_cost <num>	qalys <num>
1:	0	0.000000	0.000000	0.000000	0.0000000000
2:	1	-5.048544	22.07767	17.029126	0.0000776699
3:	2	-14.663022	24.18908	9.526063	0.0002473372
4:	3	-31.135572	26.38542	-4.750147	0.0005493339
5:	4	-53.170051	28.20682	-24.963227	0.0011979295
6:	5	-71.629879	30.12181	-41.508064	0.0018880165
7:	6	-83.857149	32.19878	-51.658373	0.0025496290
8:	7	-93.289010	33.96318	-59.325826	0.0031822142
9:	8	-110.182368	35.67620	-74.506166	0.0037537465
10:	9	-133.788003	37.44662	-96.341378	0.0044235948

iABC minus usual care, added up year by year. The intervention cost is paid in the first year after each patient's entry, so it rises most in cycle 1 and then a little each year as later entrants reach their first year. The care costs saved through avoided strokes and bleeds build up over the years:

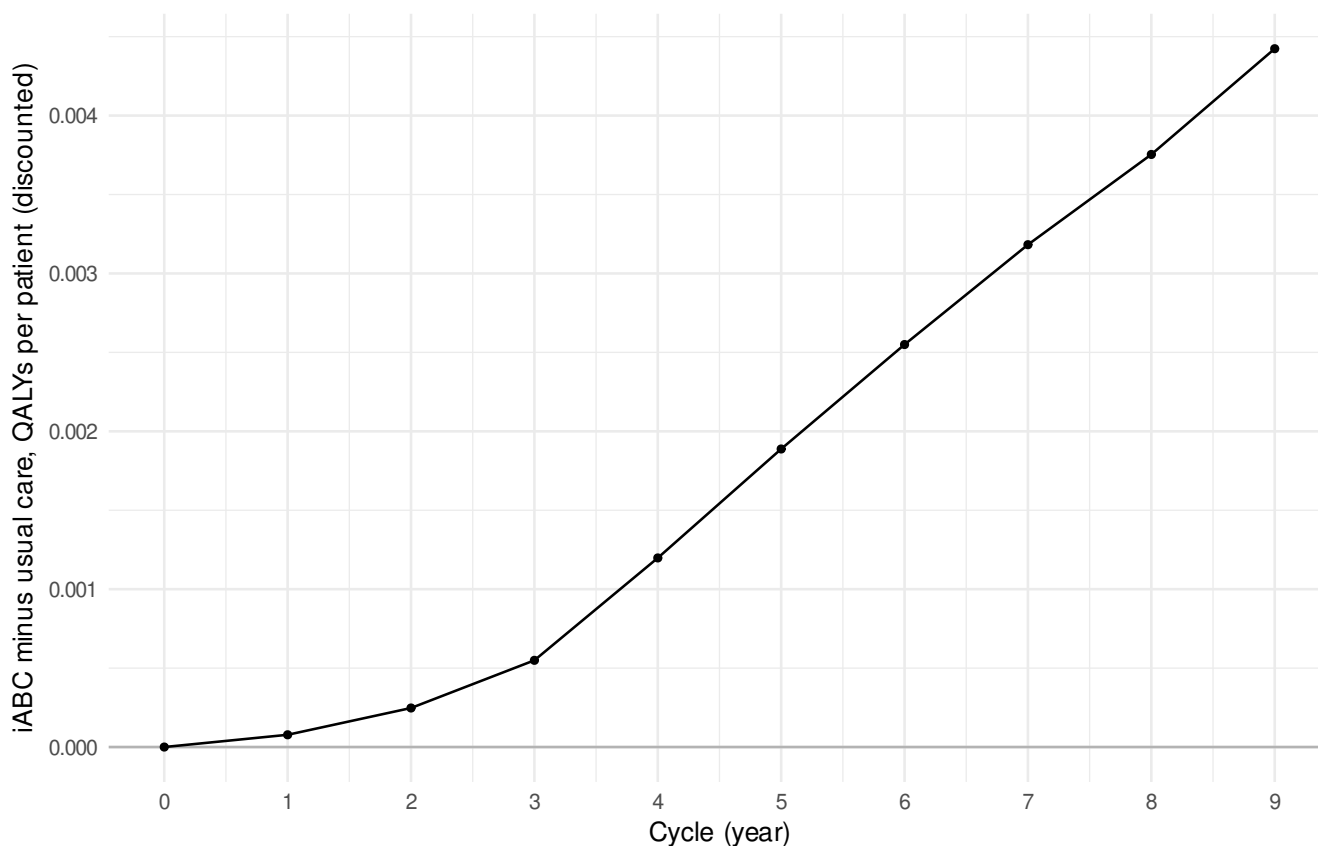
```

cost_parts <- melt(difference, id.vars = "cycle", measure.vars = c("care_cost",
↪ "intervention_cost", "total_cost"))
cost_parts[, variable := factor(variable, labels = c("Care costs", "Intervention cost",
↪ "Total"))]
ggplot(cost_parts, aes(cycle, value, colour = variable)) +
  geom_hline(yintercept = 0, colour = "grey70") + geom_line() + geom_point(size = 1) +
  scale_x_continuous(breaks = 0:9) +
  labs(x = "Cycle (year)", y = "iABC minus usual care, per patient (£, discounted)", colour =
↪ NULL) +
  theme_minimal(base_size = 10) + theme(legend.position = "top")

```



```
ggplot(difference, aes(cycle, qalys)) +
  geom_hline(yintercept = 0, colour = "grey70") + geom_line() + geom_point(size = 1) +
  scale_x_continuous(breaks = 0:9) +
  labs(x = "Cycle (year)", y = "iABC minus usual care, QALYs per patient (discounted)") +
  theme_minimal(base_size = 10)
```



4.5 The run's summary and comparison

`tm_summarise()` adds up each arm, and `tm_compare()` takes iABC minus usual care:

```
one_run <- tm_summarise(out, death, settings)
one_run
```

	arm	patients	cost_total	qaly_total	intervention_cost_total
	<char>	<int>	<num>	<num>	<num>
1:	iabc	5000	65618011	17969.23	187233.1
2:	usual_care	5000	66099718	17947.12	0.0
	cost_per_patient	qaly_per_patient	intervention_cost_per_patient	ischaemic_stroke	
	<num>	<num>	<num>	<int>	
1:	13123.60	3.593847	37.44662	450	
2:	13219.94	3.589423	0.00000	483	
	haemorrhagic_stroke	major_bleed	deaths		
	<int>	<int>	<int>		
1:	123	149	1845		
2:	133	150	1846		

```
compared <- tm_compare(copy(one_run)[, `:=`(run = 1L, seed = seed)])
compared$overall
```

	delta_cost	delta_intervention_cost	delta_qaly	icer	quadrant
	<num>	<num>	<num>	<num>	<char>
1:	-96.34138	37.44662	0.004423595	-21778.98	dominant
	delta_ischaemic_stroke	delta_haemorrhagic_stroke	delta_major_bleed	delta_deaths	
	<num>	<num>	<num>	<num>	
1:	-33	-10	-1	-1	

The incremental cost and QALYs are the last row of the year-by-year differences above:

```
difference[.N, .(total_cost, qalys)]
```

```
total_cost      qalys
  <num>         <num>
1: -96.34138 0.004423595
```

4.6 Subgroups

The discounted cost and QALYs of each patient, and the difference between the arms, by sex, by age group at entry and by cluster:

```
per_patient <- out[, .(cost = sum(cost_disc), qalys = sum(qaly_disc)), by = .(id, arm)][cohort,
  ↪ on = "id"]
per_patient <- dcast(per_patient, id + sex + agegrp + mm_cluster ~ arm, value.var = c("cost",
  ↪ "qalys"))
per_patient[, `:=`(delta_cost = cost_iabc - cost_usual_care, delta_qalys = qalys_iabc -
  ↪ qalys_usual_care)]
per_patient[, .(patients = .N, qalys_usual_care = mean(qalys_usual_care), delta_cost =
  ↪ mean(delta_cost),
  delta_qalys = mean(delta_qalys)), keyby = sex]
```

```
Key: <sex>
  sex patients qalys_usual_care delta_cost delta_qalys
  <char>   <int>         <num>         <num>         <num>
1:   men    2052          3.565699    -55.28531    0.005192154
2:  women    2948          3.605936   -124.91908    0.003888627
```

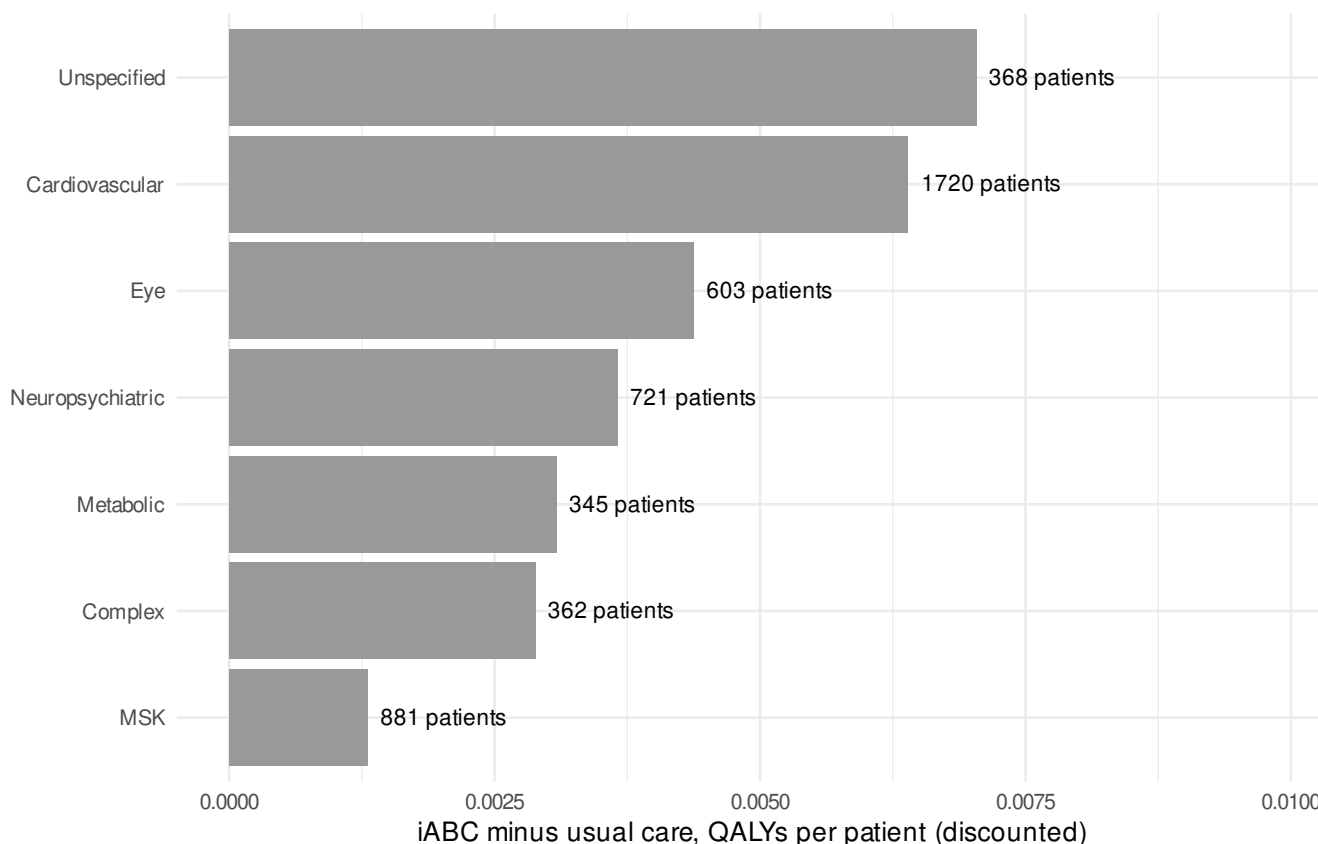
```
per_patient[, .(patients = .N, qalys_usual_care = mean(qalys_usual_care), delta_cost =
  ↪ mean(delta_cost),
  delta_qalys = mean(delta_qalys)), keyby = agegrp]
```

```
Key: <agegrp>
  agegrp patients qalys_usual_care delta_cost delta_qalys
  <char>   <int>         <num>         <num>         <num>
1: 65-69    1182          3.621257    -64.83610    0.002581692
2: 70-74     914          4.173160   -91.00585    0.003174663
3: 75-79    1010          4.035495   -125.91163    0.004061984
4: 80-84     935          3.458156   -111.88974    0.007025705
5: 85-89     630          2.874727   -94.20818    0.007203156
6: 90-94     267          2.274542  -121.31857    0.003577324
7: 95-99      62          2.014755    26.44476    0.000000000
```

```
by_cluster <- per_patient[, .(patients = .N, qalys_usual_care = mean(qalys_usual_care),
  ↪ delta_cost = mean(delta_cost),
  delta_qalys = mean(delta_qalys)), keyby = mm_cluster]
by_cluster
```

```
Key: <mm_cluster>
  mm_cluster patients qalys_usual_care delta_cost delta_qalys
  <char>   <int>         <num>         <num>         <num>
1: Cardiovascular    1720          3.633256   -78.93994    0.006389477
2:      Complex       362          2.781613  -130.78397    0.002882857
3:      Eye         603          3.686695  -130.65498    0.004377813
4:      MSK         881          3.958040   -22.70521    0.001308159
5:      Metabolic    345          3.760879  -135.70648    0.003085344
6: Neuropsychiatric   721          3.182828  -104.69996    0.003658971
7:      Unspecified   368          3.773203  -210.57268    0.007036981
```

```
ggplot(by_cluster, aes(reorder(mm_cluster, delta_qalys), delta_qalys)) +
  geom_col(fill = "grey60") + geom_text(aes(label = paste(patients, "patients")), hjust = -0.1,
    size = 3) +
  coord_flip() + expand_limits(y = max(by_cluster$delta_qalys) * 1.4) +
  labs(x = NULL, y = "iABC minus usual care, QALYs per patient (discounted)") +
  theme_minimal(base_size = 10)
```



These come from one run. In small groups, a handful of avoided events moves the average a lot, so subgroup results from a single run vary more than the overall result.

5 The full model: `tm_run()`

`tm_run()` does everything above `n_runs` times. Run r uses seed `seed + r - 1`, so its first run is the run followed in this tutorial. It returns the mean over runs and writes the results to `outputs/`:

```
results <- tm_run()
results
```

<tm_results> mean per arm over 100 runs (discounted costs and QALYs)

	arm	patients	cost_total	qaly_total	intervention_cost_total
	<char>	<num>	<num>	<num>	<num>
1:	iabc	5000	65493591	18004.50	187494.6
2:	usual_care	5000	66035435	17980.58	0.0
	cost_per_patient	qaly_per_patient	intervention_cost_per_patient	ischaemic_stroke	
	<num>	<num>	<num>	<num>	
1:	13098.72	3.601	37.499	429.23	
2:	13207.09	3.596	0.000	461.93	

	haemorrhagic_stroke	major_bleed	deaths
	<num>	<num>	<num>
1:	121.54	147.47	1850.85
2:	129.50	160.45	1853.00

iABC minus usual care (mean over runs, per patient):

Incremental cost: €-108.37 (intervention cost: €37.5)

Incremental QALYs: 0.00478

ICER: iABC dominant (cheaper, more QALYs)

Difference in events (whole cohort): ischaemic_stroke -32.7, haemorrhagic_stroke -8.0, major_bleed -13.0, deaths -2.1

Written to outputs/: summary.csv, summary_by_run.csv, comparison.csv, comparison_by_run.csv, ce_plane.png

The first run of `tm_run()` and the run followed above, side by side:

```

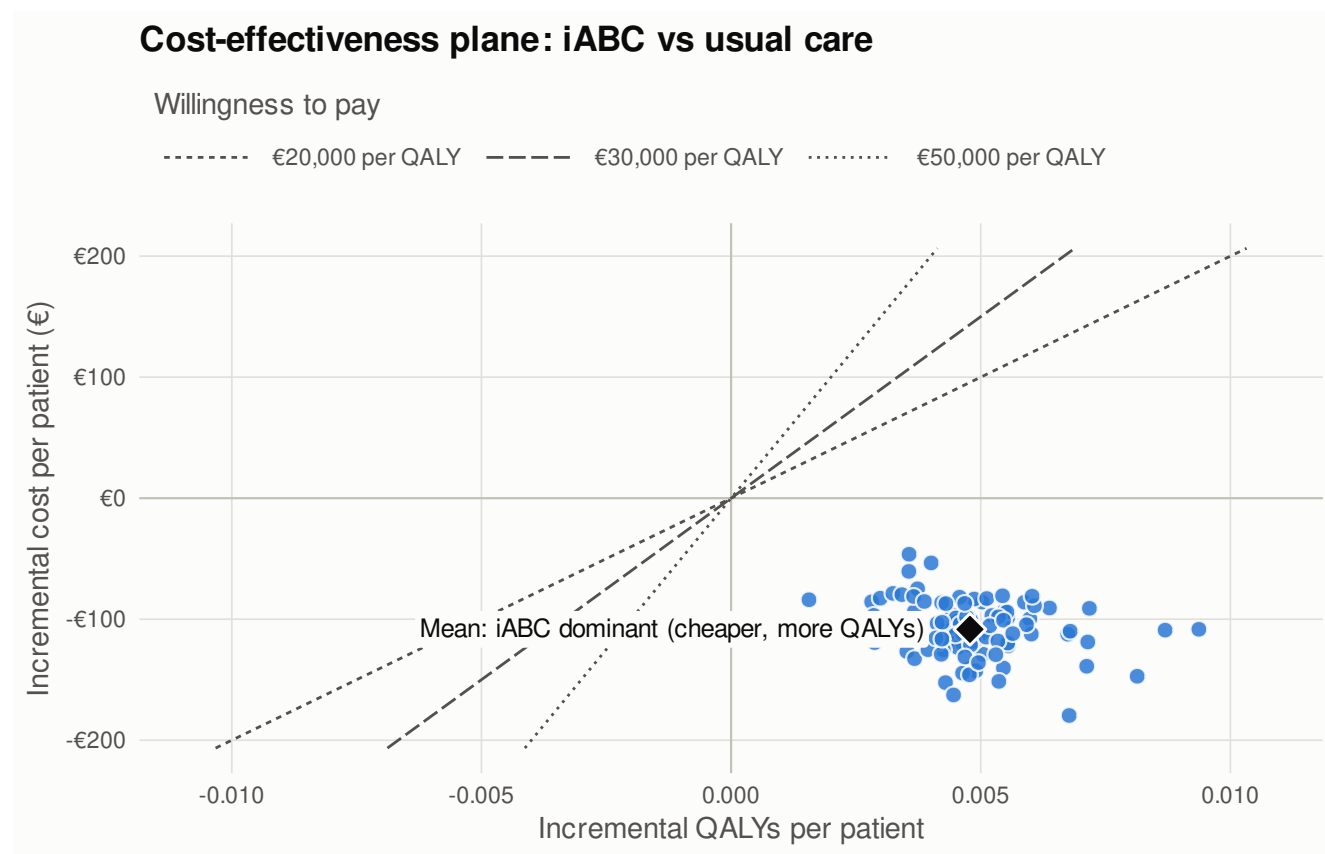
rbind(cbind(source = "step by step", one_run[, .(arm, cost_per_patient, qaly_per_patient,
  ↪ deaths)]),
      cbind(source = "tm_run(), run 1", results$by_run[run == 1, .(arm, cost_per_patient,
  ↪ qaly_per_patient, deaths)]))

```

	source	arm	cost_per_patient	qaly_per_patient	deaths
	<char>	<char>	<num>	<num>	<int>
1:	step by step	iabc	13123.60	3.593847	1845
2:	step by step	usual_care	13219.94	3.589423	1846
3:	tm_run(), run 1	iabc	13123.60	3.593847	1845
4:	tm_run(), run 1	usual_care	13219.94	3.589423	1846

Each point on the cost-effectiveness plane is one run; the diamond is the mean across runs:

```
plot(results)
```



```
list.files("outputs")
```

```
[1] "ce_plane.png"          "comparison_by_run.csv" "comparison.csv"  
[4] "summary_by_run.csv"    "summary.csv"
```